3) Linear Methods for Regression

Assume we have a dataset with with 3 predictors (x_1, x_2, x_3) and 1 dependent variable y:

x_1	x_2	x_3	y
1	2	1	3
3	1	5	2
7	1	2	3

1 sample

1 predictor

We write the input vector as:

$$X = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 5 \\ 7 & 1 & 2 \end{bmatrix}$$ Matrix Form

Similarly, the output vector is $Y = \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}$

If we have p predictors and n datasamples, then, X has dimension $n \times p$ (n rows and p columns). Y has dimension $n \times 1$.

3.2) Linear Regression And Least Squares

The linear regression model, has form

$$Y = X\beta + \varepsilon \iff \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_{n\times 1} = \underbrace{\begin{bmatrix} 1 & x_{11} & x_{21} & \cdots & x_{p1} \\ 1 & x_{12} & x_{22} & \cdots & x_{p2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1n} & x_{2n} & \cdots & x_{kn} \end{bmatrix}}_{n\times k} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}}_{k\times 1} + \underbrace{\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}}_{n\times 1}$$

For a single observation, this becomes:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$$

The linear regression model has high bias as it assumes that $\mathbb{E}[Y|X]$ is linear (or assuming linearity is reasonable).

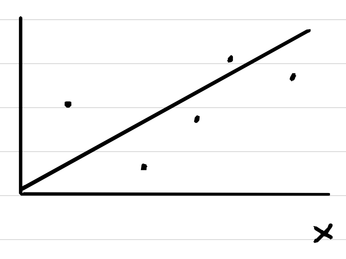
To estimate $\beta = [\beta_0, \beta_1, \dots, \beta_p]^T$ we use a set of training data with N observations $(x_1, y_1), \dots, (x_N, y_N)$:

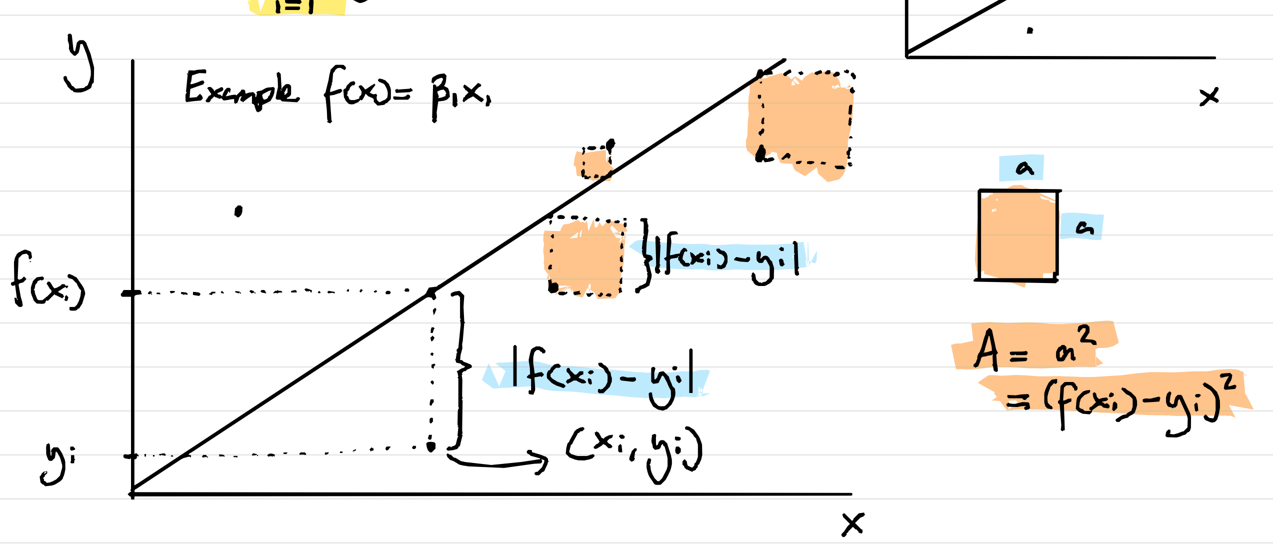
- $x_i \in \mathbb{R}^p$, $x_i = [x_{i1}, x_{i2}, \dots, x_{ip}]$
- $y_i \in \mathbb{R}$

Let f be our linear model: $f: \mathbb{R}^p \longrightarrow \mathbb{R}$, $x_i \longmapsto y_i$

Then, the least squares method minimises the residual sum of squares:

$$RSS(\beta) = \sum_{i=1}^{N} (y_i - f(x_i))^2$$





In matrix form:

$$RSS(\beta) = (Y - X\beta)^T (Y - XB)$$

The minimum is obtained by setting:

$$\frac{\partial RSS}{\partial \beta} = 0 \implies -2X^TY + 2X^TX\beta = 0$$

$$\implies X^TX\beta = X^TY$$

OLS Solution $\implies \boxed{\hat{\beta} = (X^TX)^{-1}X^TY}$

Worth giving this a deeper look

*) Assuming that X has full column rank, hence X^TX is invertible and positive definite.

$\hat{Y} = X\hat{\beta}$ * ^ denotes an estimate

$= X(X^TX)^{-1}X^TY$ (From OLS)

$= HY$ (Letting $H = X(X^TX)^{-1}X^T$)

H is sometimes called the hat matrix. Do you see why?

If the columns of X are not linearly independent (e.g. $x_1 = 2x_2$), then $\hat{\beta}$ is not uniquely defined. In this case, the OLS formula does not hold. This also happens if $p > N$.

The variance of the estimator is:

$$Var(\hat{\beta}) = (X^TX)^{-1}\sigma^2, \qquad \hat{\sigma}^2 = \frac{1}{N-p-1}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

$\hat{\sigma}^2$ is an unbiased estimate of σ^2. If we assume the conditional expectation of Y ($\mathbb{E}[Y|X]$) is linear on $X_1, \dots, X_p$, and Gaussian errors, then.

$$Y = \mathbb{E}(Y|X_1, \dots, X_p) + \varepsilon$$

$$= \beta_0 + \sum_{j=1}^{p} X_j \beta_j + \varepsilon \qquad \varepsilon \sim N(0, \sigma^2)$$

Then, $\hat{\beta} \sim N(\beta, (X^TX)^{-1}\sigma^2) \implies$ This can be used to construct hypothesis tests and confidence intervals for β_j

The test hypothesis for a particular coefficient $\beta_j = 0$.

Z-score: $z_j = \left|\frac{\hat{\beta}_j}{\hat{\sigma}\sqrt{v_j}}\right|$ v_j: j-th diagonal element of $(X^TX)^{-1}$

z_j is distributed as t_{N-p-1}. If σ is known (not estimated), then z_j is normally distributed. For $N-p-1 > 100$ $t_{N-p-1} \sim N$.

F-statistic

$$F = \frac{\frac{RSS_0 - RSS_1}{p_1 - p_0}}{\frac{RSS_1}{n - p_1}}$$

RSS_1: bigger model
RSS_0: smaller model
p_1 & p_0: n° of predictor (β_0 counts)
N: n° of training examples

Example

Baseline (model 0): $F(x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_4 x_4$

- $RSS_0 = 32.81$ (Residual sum of squares on data)

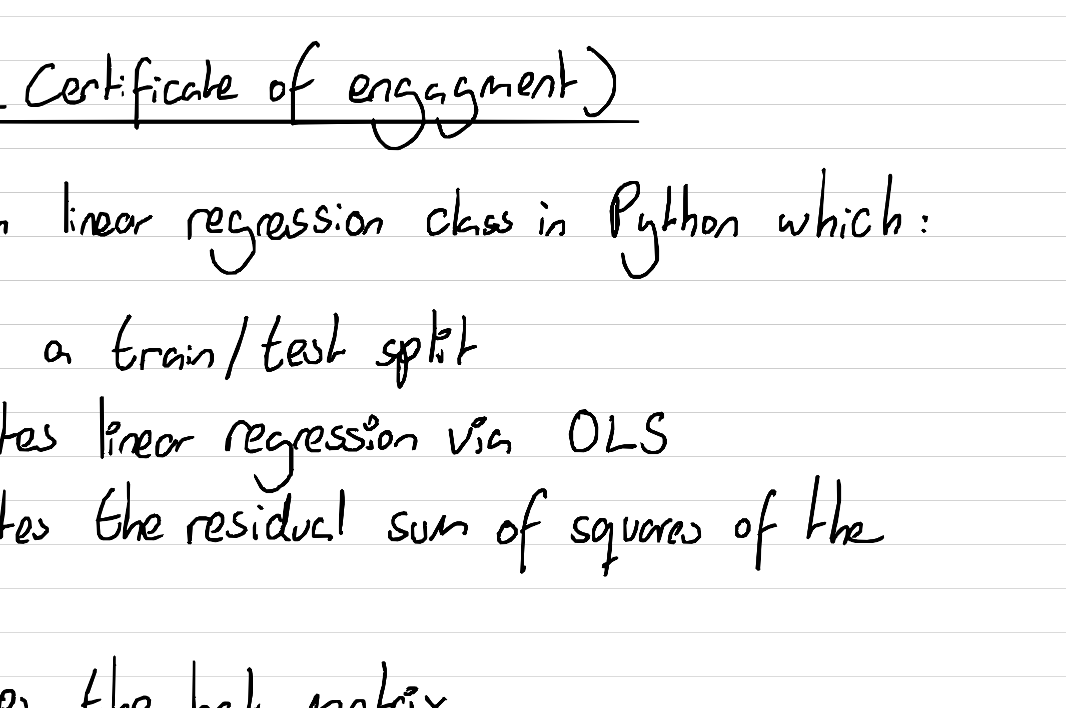
New (model 1): $F(x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_8 x_8$ (add 4 predictors)

- $RSS_1 = 29.43$

If both models are trained on 67 data examples, then:

$$F = \frac{(32.81 - 29.43)/(9-5)}{29.43/(67-9)} = 1.67$$

$P(F_{4,58} > 1.67) = 0.17 \implies$ Not significant



Exercises (Certificate of engagment)

1) Create a linear regression class in Python which:

- Creates a train/test split
- Computes linear regression via OLS
- Computes the residual sum of squares of the model
- Computes the hat matrix
- Computes the variance of the estimate $\hat{\beta}$
- Computes the z-scores of $\hat{\beta}$

2) Create a class to compute the F-statistic between two linear regression models.

3) Use it in the Kaggle playground competition.